F16AG426-VS
VARIABLE SPEED DRIVE

SYSTEM INTERCONNECTION DIAGRAM 2000-EL005-B02-10001

RD WH BU
11,000V
FCF 100A
QAC 100A 100A 400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE 400A
EARTH BAR
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10016 SHEET 14

L1 L2 L3
LINE SIDE INCOMING SUPPLY SHEET 6

OUTGOING MOTOR SUPPLY SHEET 6

T1 T2 T3
CABLE SCREEN EARTHED
M 3~
F16-AG-426
ROUGHER FLOTATION CELL 3 AGITATOR
3,300V
500 kW

F16-AG-426
MOTOR HEATER TERMINAL BOX
F16-HT-546
X2 54 53
F16HT546-L
F16HT546-N
0.3kW
ANTI-CONDENSATION HEATER

MOTOR SPACE HEATER CIRCUIT SHEET 6

X81 (PART OF)
RTD TEMPERATURE MONITOR RELAY
SHEET 6

F16-AG-426
MOTOR RTD TERMINAL BOX

X81
PT100 RTD INPUT CH1: 13 (+) 14 (COM) 15 (-)
PT100 RTD INPUT CH2: 16 (+) 17 (COM) 18 (-)
PT100 RTD INPUT CH3: 19 (+) 20 (COM) 21 (-)
PT100 RTD INPUT CH4: 22 (+) 23 (COM) 24 (-)
PT100 RTD INPUT CH5: 25 (+) 26 (COM) 27 (-)
PT100 RTD INPUT CH6: 28 (+) 29 (COM) 30 (-)
PT100 RTD INPUT CH7: 31 (+) 33 (COM) 34 (-)
PT100 RTD INPUT CH8: 35 (COM) 36 (-)

F16AG426A-RD
F16AG426A-WH
F16AG426A-BK
F16AG426B-RD
F16AG426B-WH
F16AG426B-BK
F16AG426C-RD
F16AG426C-WH
F16AG426C-BK
F16AG426D-RD
F16AG426D-WH
F16AG426D-BK
F16AG426E-RD
F16AG426E-WH
F16AG426E-BK

RTD
MOTOR WINDING RTD (RD PHASE) (TD1)
MOTOR WINDING RTD (WH PHASE) (TD3)
MOTOR WINDING RTD (BU PHASE) (TD5)
MOTOR BEARING RTD (DRIVE END) (BD10)
MOTOR BEARING RTD (NON DRIVE END) (BD20)

F16-SW-011 11kV FEEDER START/STOP CONTROL
2000-EL003-F16-10016 SHEET 7 3A & 3B
X3 3 1139
X3 17 3703
X2 67
X2 68
K41 K41
MV CONTACTOR ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD HEALTHY
2000-EL003-F16-10016 SHEET 10 4A
X2 4 1179
X2 25 1199
X2 63
X2 64
K41 K41 K41
MV INPUT ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD TRIP
2000-EL003-F16-10016 SHEET 10 3A
X2 3 1219
X2 24 1239
X2 65
X2 66
K41 K41 K41
MV INPUT TRIP CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER ARC FLASH TRIP
2000-EL003-F16-10016 SHEET 8 5A & 5D
X2 56 1259
X2 44 1279
X2 70
X2 69
A30 (PART OF)
X2 15 TRIP X2 16
ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F16-DB-101
2000-F16-TLD-EL-10054
RD WH BU BK
R15 W15 B15
25A
415VAC
EARTH BAR
F16DB101-R15
F16DB101-W15
F16DB101-B15
F16DB101-BK15
X4 1 2 3 4 PE1
VSD AUXILIARY POWER SUPPLY SHEET 6

F16-DB-201
2000-F16-TLD-EL-10057
BU BK
B4
10A
240VAC
EARTH BAR
F16DB201-B4
F16DB201-BK4
X4 5 6 PE2
VSD UPS POWER SUPPLY SHEET 6

TO FLOTATION 11kV SWITCHROOM
F16-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F16-ETH-001
2000-F16-TER-IC-10002
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47
ETHERNET ADAPTOR RJ45
A2 (PART OF)
DI-2B 9

SAFE TORQUE OFF CIRCUIT SHEET 6
X86 1 2 3 4
F16AG426VS-1409
F16AG426VS-1419
F16AG426VS-1429
F16AG426VS-1439

F16AG426-LCS
LOCAL CONTROL STATION
X1 1 2 3 4
LOS

X2 13 14
F16AG426VS-1459
F16AG426VS-1469
24VDC CONTROL SHEET 6
X1 5 6 E
START

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F16AG426-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.

LEGEND

- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL

PANEL A07

2000-F16-SCM-EL-10113_Rev1

F16-SW-011 F16-AG-426 scavenger flot. cell 3 agi VSD feeder schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCOM GROWTH PROJECT\03 DELIVERABLES\EL\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SCM-EL-10113.DWG 09SEP25 6:39AM HOTSEAT04